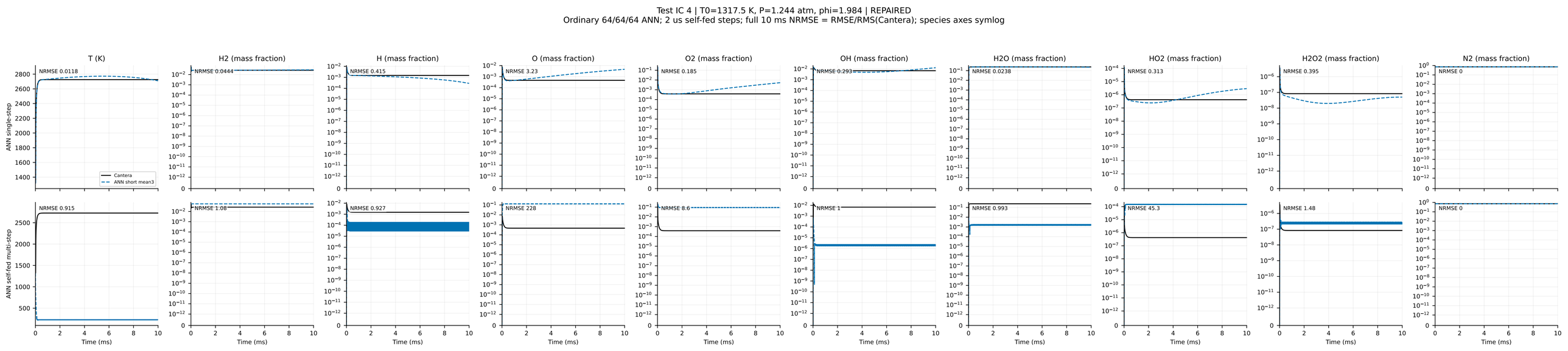
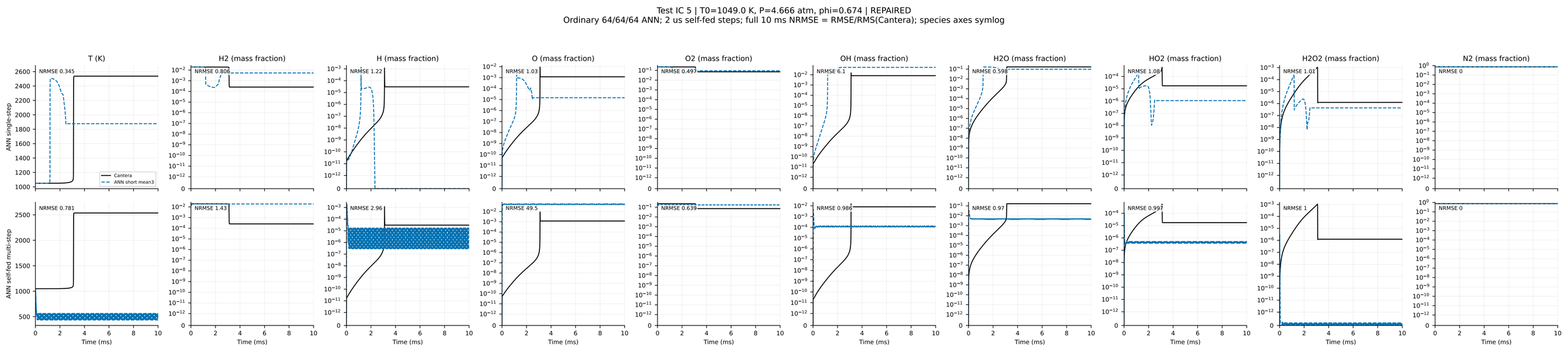
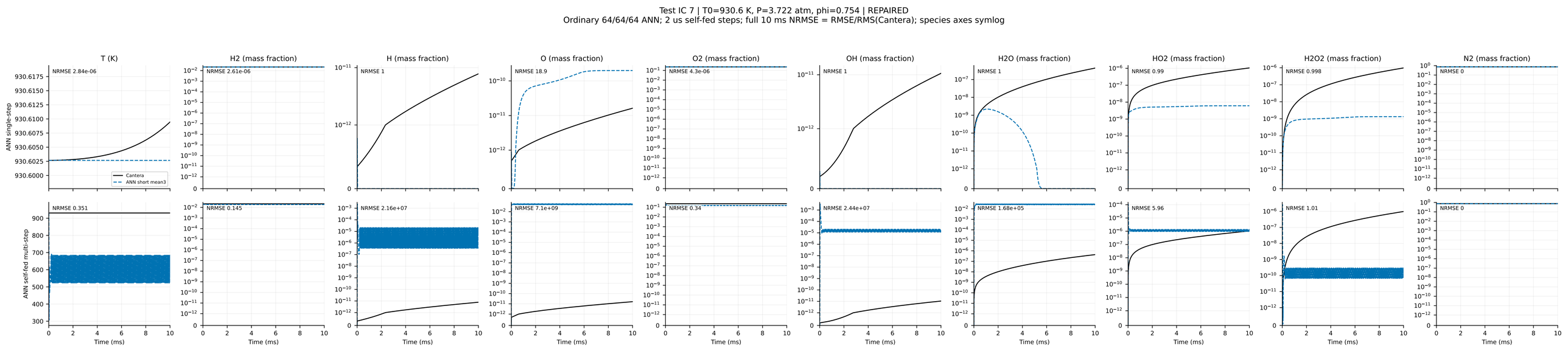
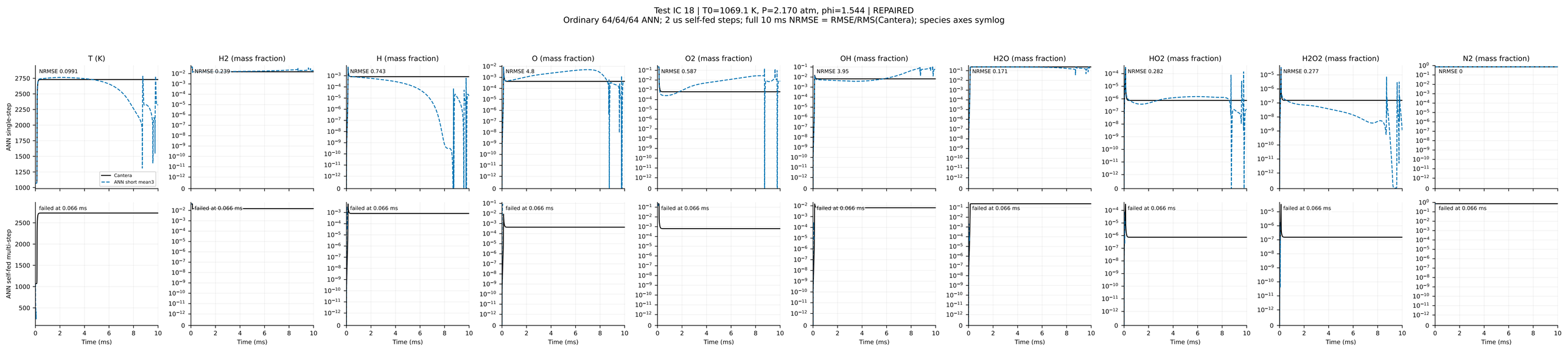
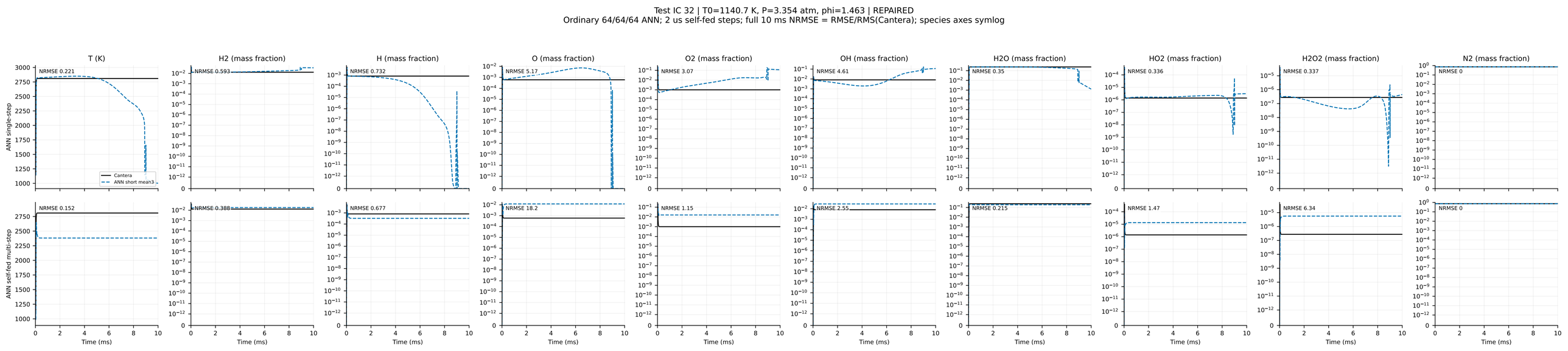
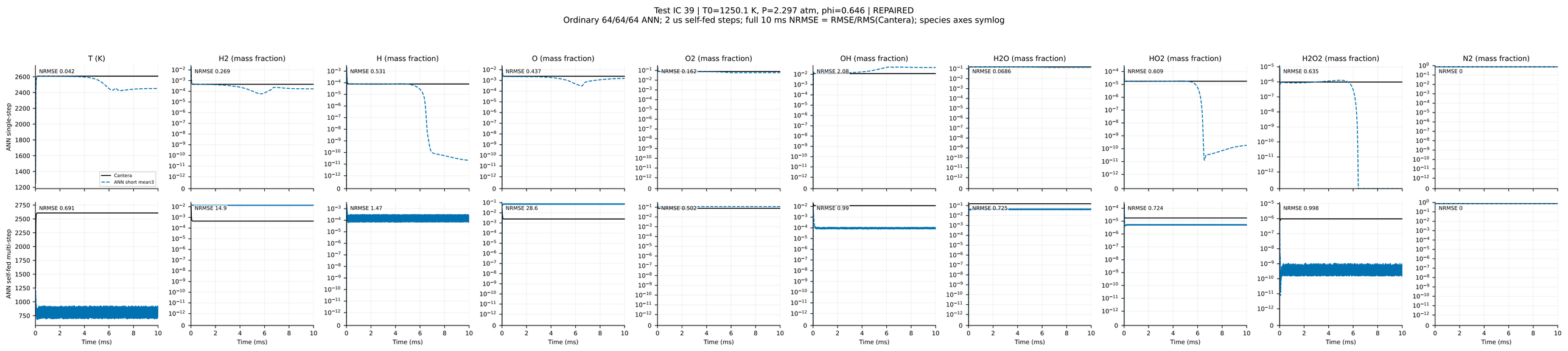

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0118
NRMSE 0.0444
NRMSE 0.415
NRMSE 3.23
NRMSE 0.185
NRMSE 0.293
NRMSE 0.0238
NRMSE 0.313
NRMSE 0.395
NRMSE 0
NRMSE 0.915
NRMSE 1.08
NRMSE 0.927
NRMSE 228
NRMSE 8.6
NRMSE 1
NRMSE 0.993
NRMSE 45.3
NRMSE 1.48
NRMSE 0
Cantera
ANN short mean3
Time (ms)

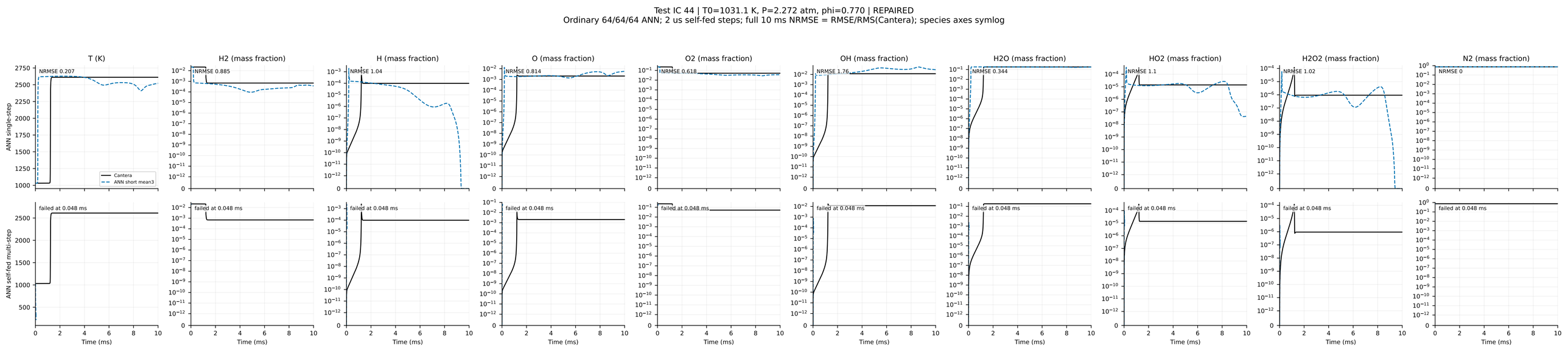

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.345
NRMSE 0.806
NRMSE 1.22
NRMSE 1.03
NRMSE 0.497
NRMSE 6.1
NRMSE 0.598
NRMSE 1.08
NRMSE 1.01
NRMSE 0
NRMSE 0.781
NRMSE 1.43
NRMSE 2.96
NRMSE 49.5
NRMSE 0.639
NRMSE 0.986
NRMSE 0.97
NRMSE 0.997
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

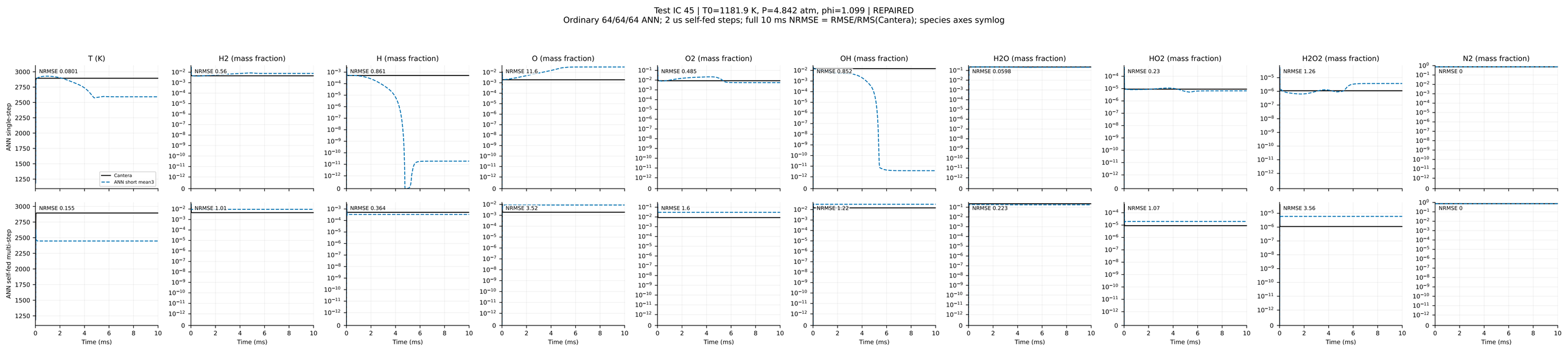

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 2.84e-06
NRMSE 2.61e-06
NRMSE 1
NRMSE 18.9
NRMSE 4.3e-06
NRMSE 1
NRMSE 1
NRMSE 0.99
NRMSE 0.998
NRMSE 0
NRMSE 0.351
NRMSE 0.145
NRMSE 2.16e+07
NRMSE 7.1e+09
NRMSE 0.34
NRMSE 2.44e+07
NRMSE 1.68e+05
NRMSE 5.96
NRMSE 1.01
NRMSE 0
Cantera
ANN short mean3
Time (ms)

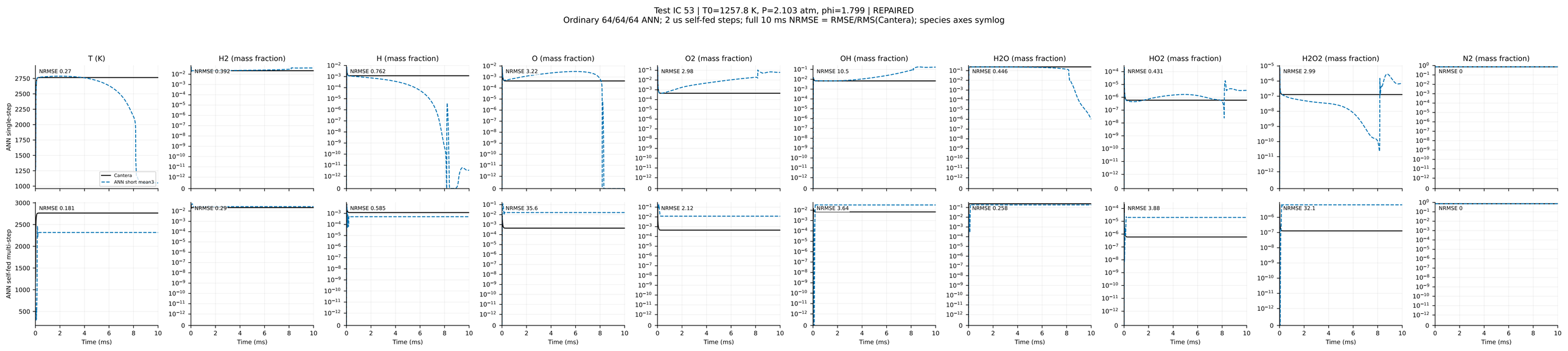

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0991
NRMSE 0.239
NRMSE 0.743
NRMSE 4.8
NRMSE 0.587
NRMSE 3.95
NRMSE 0.171
NRMSE 0.282
NRMSE 0.277
NRMSE 0
Cantera
ANN short mean3
failed at 0.066 ms
Time (ms)

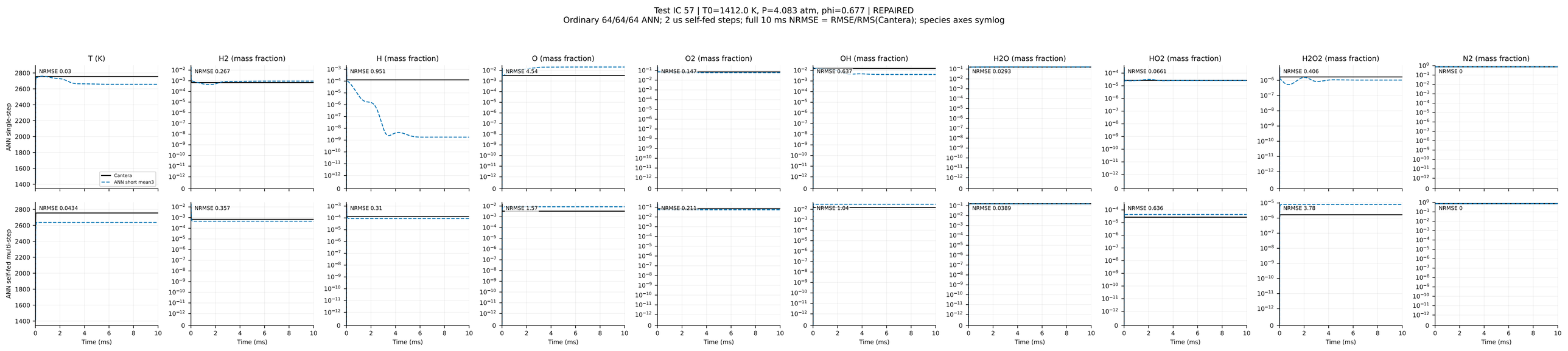

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.221
NRMSE 0.593
NRMSE 0.732
NRMSE 5.17
NRMSE 3.07
NRMSE 4.61
NRMSE 0.35
NRMSE 0.336
NRMSE 0.337
NRMSE 0
NRMSE 0.152
NRMSE 0.388
NRMSE 0.677
NRMSE 18.2
NRMSE 1.15
NRMSE 2.55
NRMSE 0.215
NRMSE 1.47
NRMSE 6.34
NRMSE 0
Cantera
ANN short mean3
Time (ms)

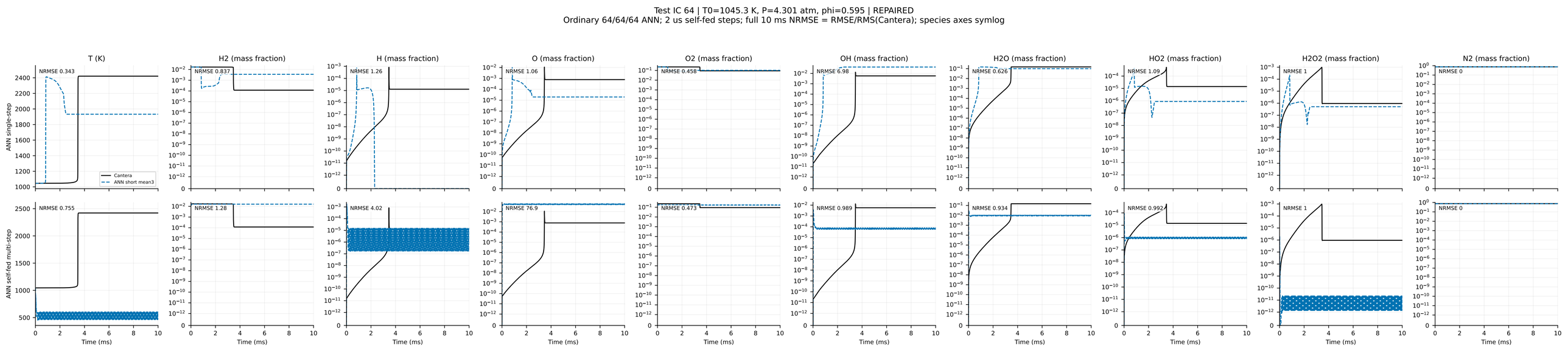

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.042
NRMSE 0.269
NRMSE 0.531
NRMSE 0.437
NRMSE 0.162
NRMSE 2.08
NRMSE 0.0686
NRMSE 0.609
NRMSE 0.635
NRMSE 0
NRMSE 0.691
NRMSE 14.9
NRMSE 1.47
NRMSE 28.6
NRMSE 0.502
NRMSE 0.99
NRMSE 0.725
NRMSE 0.724
NRMSE 0.998
NRMSE 0
Cantera
ANN short mean3
Time (ms)

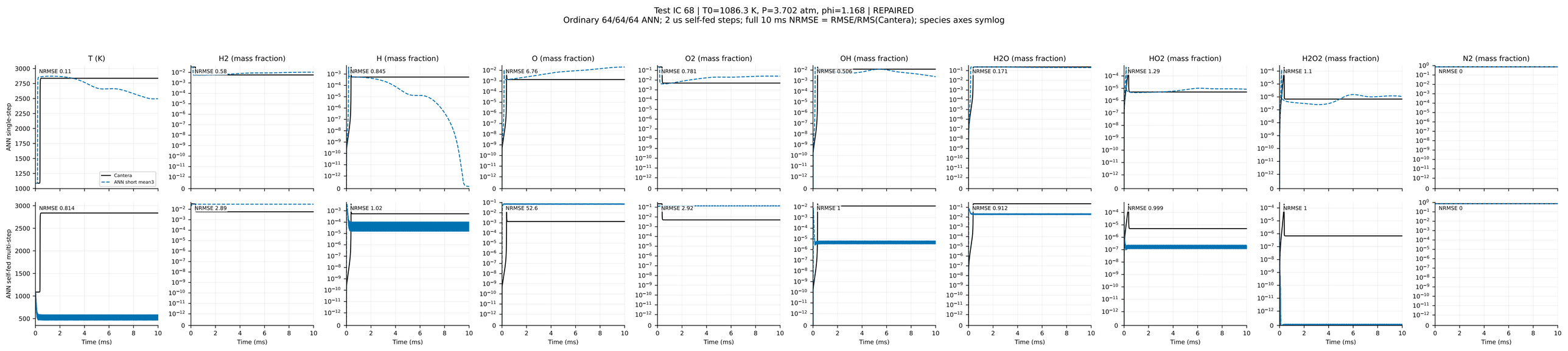

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.207
NRMSE 0.885
NRMSE 1.04
NRMSE 0.814
NRMSE 0.618
NRMSE 1.76
NRMSE 0.344
NRMSE 1.1
NRMSE 1.02
NRMSE 0
Cantera
ANN short mean3
failed at 0.048 ms
Time (ms)

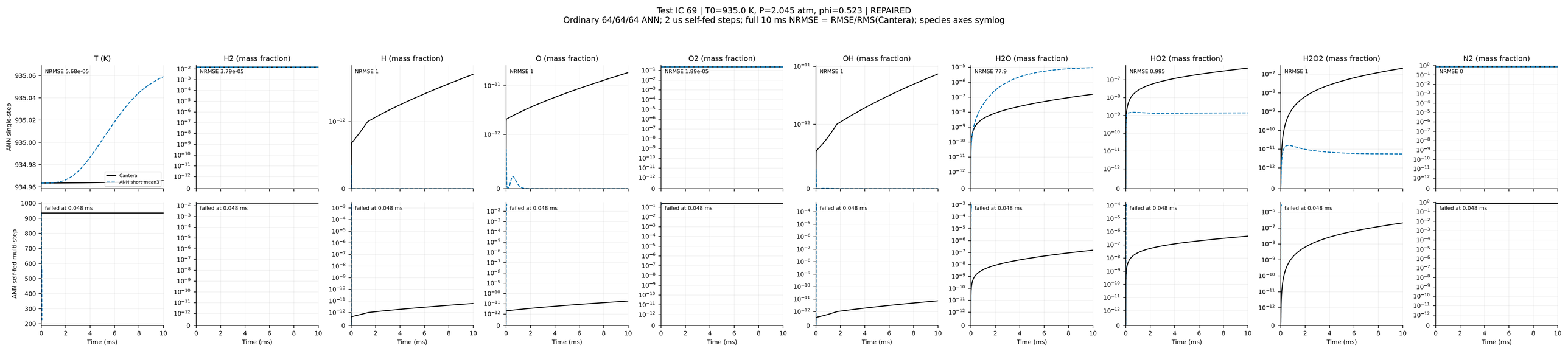

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0801
NRMSE 0.56
NRMSE 0.861
NRMSE 11.6
NRMSE 0.485
NRMSE 0.852
NRMSE 0.0598
NRMSE 0.23
NRMSE 1.26
NRMSE 0
NRMSE 0.155
NRMSE 1.01
NRMSE 0.364
NRMSE 3.52
NRMSE 1.6
NRMSE 1.22
NRMSE 0.223
NRMSE 1.07
NRMSE 3.56
NRMSE 0
Cantera
ANN short mean3
Time (ms)

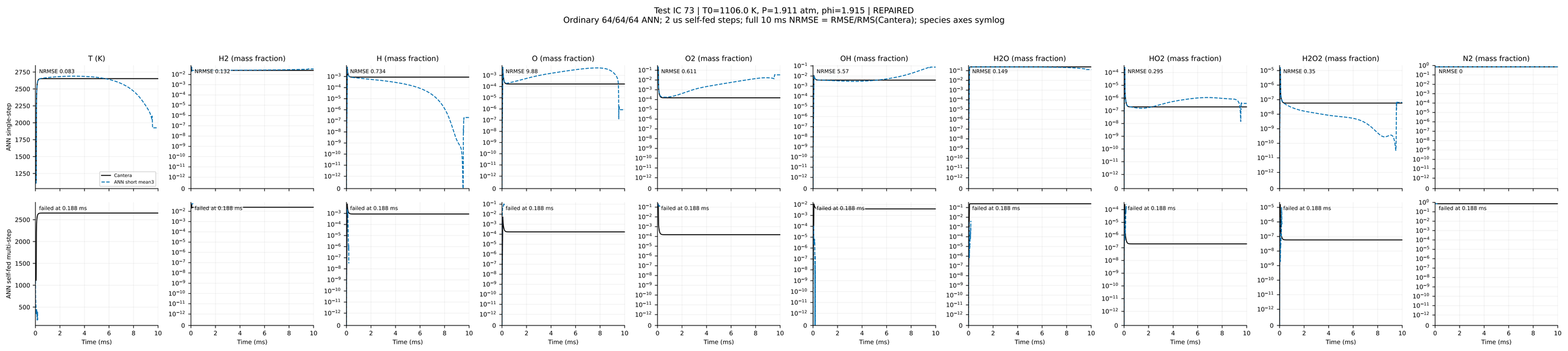

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.27
NRMSE 0.392
NRMSE 0.762
NRMSE 3.22
NRMSE 2.98
NRMSE 10.5
NRMSE 0.446
NRMSE 0.431
NRMSE 2.99
NRMSE 0
NRMSE 0.181
NRMSE 0.29
NRMSE 0.585
NRMSE 35.6
NRMSE 2.12
NRMSE 3.64
NRMSE 0.258
NRMSE 3.88
NRMSE 32.1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

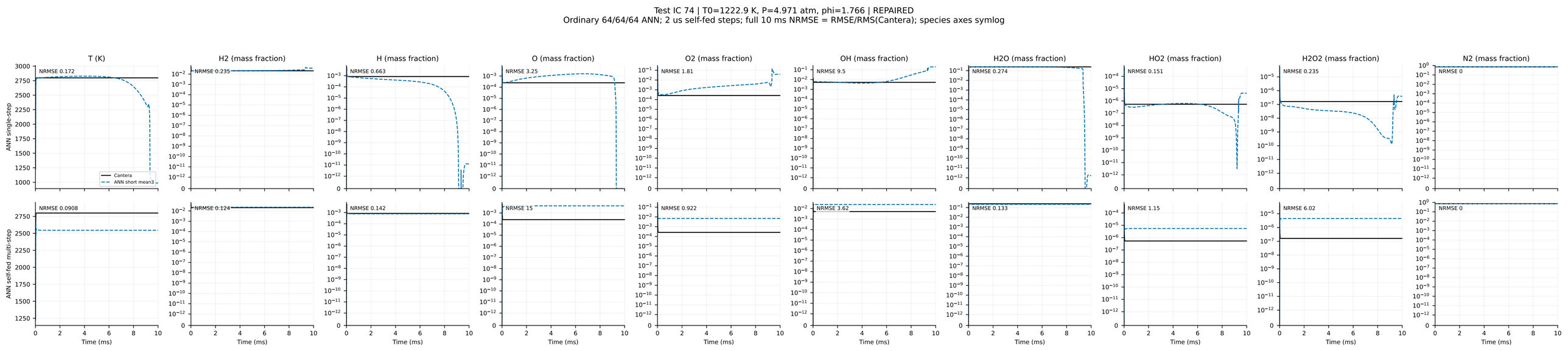

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.03
NRMSE 0.267
NRMSE 0.951
NRMSE 4.54
NRMSE 0.147
NRMSE 0.637
NRMSE 0.0293
NRMSE 0.0661
NRMSE 0.406
NRMSE 0
NRMSE 0.0434
NRMSE 0.357
NRMSE 0.31
NRMSE 1.57
NRMSE 0.211
NRMSE 1.04
NRMSE 0.0389
NRMSE 0.636
NRMSE 3.78
NRMSE 0
Cantera
ANN short mean3
Time (ms)

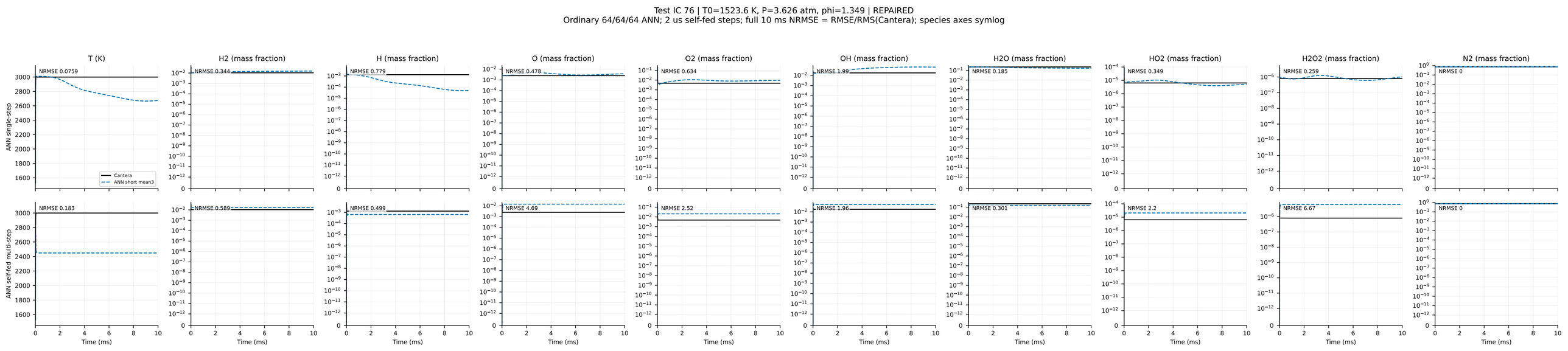

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.343
NRMSE 0.837
NRMSE 1.26
NRMSE 1.06
NRMSE 0.458
NRMSE 6.98
NRMSE 0.626
NRMSE 1.09
NRMSE 1
NRMSE 0
NRMSE 0.755
NRMSE 1.28
NRMSE 4.02
NRMSE 76.9
NRMSE 0.473
NRMSE 0.989
NRMSE 0.934
NRMSE 0.992
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

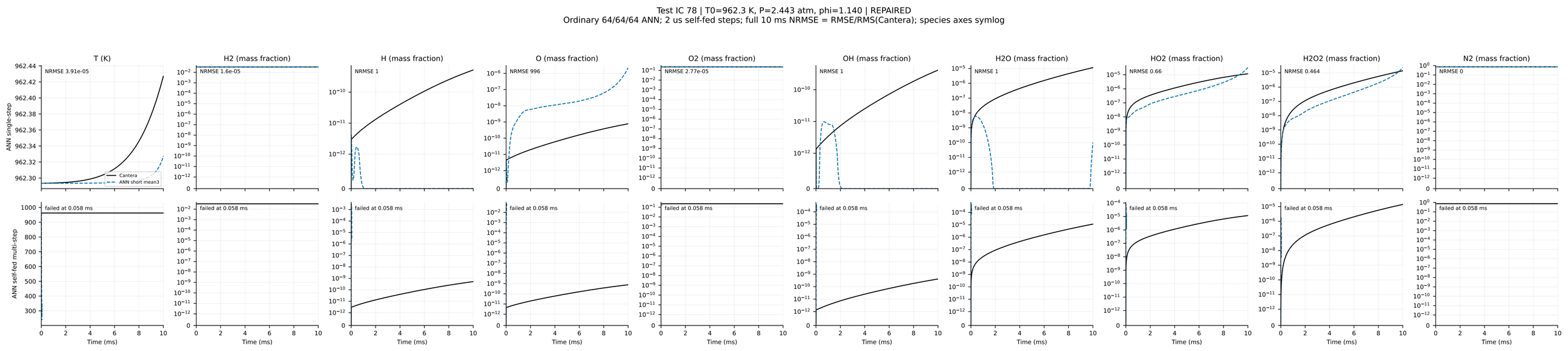

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.11
NRMSE 0.58
NRMSE 0.845
NRMSE 6.76
NRMSE 0.781
NRMSE 0.506
NRMSE 0.171
NRMSE 1.29
NRMSE 1.1
NRMSE 0
NRMSE 0.814
NRMSE 2.89
NRMSE 1.02
NRMSE 52.6
NRMSE 2.92
NRMSE 1
NRMSE 0.912
NRMSE 0.999
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

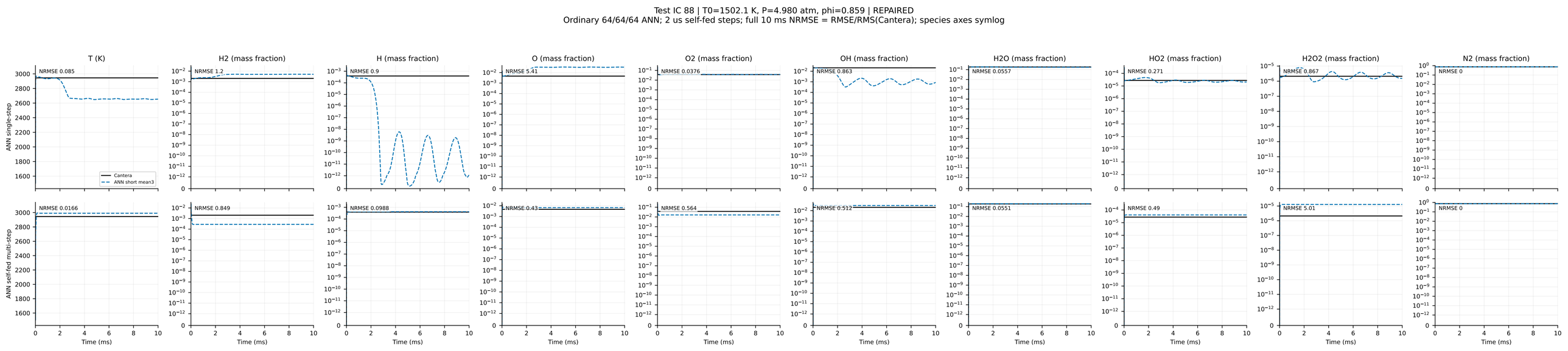

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 5.68e-05
NRMSE 3.79e-05
NRMSE 1
NRMSE 1
NRMSE 1.89e-05
NRMSE 1
NRMSE 77.9
NRMSE 0.995
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
failed at 0.048 ms
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.083
NRMSE 0.132
NRMSE 0.734
NRMSE 9.88
NRMSE 0.611
NRMSE 5.57
NRMSE 0.149
NRMSE 0.295
NRMSE 0.35
NRMSE 0
Cantera
ANN short mean3
failed at 0.188 ms
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.172
NRMSE 0.235
NRMSE 0.663
NRMSE 3.25
NRMSE 1.81
NRMSE 9.5
NRMSE 0.274
NRMSE 0.151
NRMSE 0.235
NRMSE 0
NRMSE 0.0908
NRMSE 0.124
NRMSE 0.142
NRMSE 15
NRMSE 0.922
NRMSE 3.62
NRMSE 0.133
NRMSE 1.15
NRMSE 6.02
NRMSE 0
Cantera
ANN short mean3
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0759
NRMSE 0.344
NRMSE 0.779
NRMSE 0.478
NRMSE 0.634
NRMSE 1.99
NRMSE 0.185
NRMSE 0.349
NRMSE 0.259
NRMSE 0
NRMSE 0.183
NRMSE 0.589
NRMSE 0.499
NRMSE 4.69
NRMSE 2.52
NRMSE 1.96
NRMSE 0.301
NRMSE 2.2
NRMSE 6.67
NRMSE 0
Cantera
ANN short mean3
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 3.91e-05
NRMSE 1.6e-05
NRMSE 1
NRMSE 996
NRMSE 2.77e-05
NRMSE 1
NRMSE 1
NRMSE 0.66
NRMSE 0.464
NRMSE 0
Cantera
ANN short mean3
failed at 0.058 ms
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.085
NRMSE 1.2
NRMSE 0.9
NRMSE 5.41
NRMSE 0.0376
NRMSE 0.863
NRMSE 0.0557
NRMSE 0.271
NRMSE 0.867
NRMSE 0
NRMSE 0.0166
NRMSE 0.849
NRMSE 0.0988
NRMSE 0.43
NRMSE 0.564
NRMSE 0.512
NRMSE 0.0551
NRMSE 0.49
NRMSE 5.01
NRMSE 0
Cantera
ANN short mean3
Time (ms)